

ETF SOS Depth Is Not Determined by (N, r)

An exact counterexample to Question 4.1 of arXiv:1901.10697

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Status. `candidate_full_counterexample_likely_valid`. The finite part of the proof is certified with exact integer and rational arithmetic, rather than floating-point SDP output.

Source question. Bandeira–Kunisky, arXiv:1901.10697, Question 4.1 on PDF page 12 asks for the largest even d such that the Gram matrix of a real ETF of N vectors in $\mathbb{R}^r$ belongs to the degree- d generalized elliptope $\mathcal{E}_d^N$, and asks whether this largest degree depends only on N and r .

4 Open Problems

Returning to the more general family of relaxations of the cut polytope described in Definition [1.2](#) the main question we propose for investigation is the following.

Question 4.1. *Given the Gram matrix of a real ETF of N vectors in $\mathbb{R}^r$, what is the largest d for which it belongs to $\mathcal{E}_d^N$? Does the answer depend only on N and r ?*

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Definitions. The cut polytope is

$$\mathcal{C}^N = \text{conv}\{zz^\top : z \in \{\pm 1\}^N\}.$$

The generalized elliptopes form the decreasing SOS hierarchy

$$\mathcal{E}_2^N \supseteq \mathcal{E}_4^N \supseteq \cdots \supseteq \mathcal{E}_{N+1\{N \text{ odd}\}}^N = \mathcal{C}^N. \quad (1)$$

For a graph with adjacency matrix A , its Seidel matrix is $S = \mathbf{1}\mathbf{1}^\top - I - 2A$.

Proof intuition. Strongly regular graphs with parameters $(26, 10, 3, 4)$ all have Seidel spectrum $\{5^{(13)}, -5^{(13)}\}$. Consequently $I + S/5$ is always the Gram matrix of a real $(26, 13)$ ETF. What varies with the switching class is the population of sign vectors in the $+5$ eigenspace. In the first example that population is rich enough that its uniform covariance is the ETF Gram matrix. In the second, there are only fourteen sign lines. Positivity forces any cut decomposition to use only those lines, but their unique affine representation of the Gram matrix assigns one line negative mass. This gives two identical frame parameter pairs but different terminal SOS membership.

Theorem. *There exist two real equiangular tight frames with the same parameters $(N, r) = (26, 13)$ whose Gram matrices have different largest generalized-elliptope degrees. One Gram matrix belongs to $\mathcal{E}_{26}^{26} = \mathcal{C}^{26}$, while the other does not. Hence the answer to the second sentence of Question 4.1 does not depend only on N and r .*

The two exact graphs. Let G_c, G_n be the graphs encoded by the following graph6 strings, respectively:

```
Y}qCCmMlDSIPRCXPKSQhHQooE^'E'pHQoQuHBRc_oo}EHSMILOkEWXB_
Y}qCCmMlDSIQRBWokcadAROkE]XEXPHecTJ_AlWGooveDEcISRDEX_p_
```

They are the first two entries in Brendan McKay's catalogue of the ten strongly regular graphs with parameters $(26, 10, 3, 4)$. Write A_c, A_n for their adjacency matrices and

$$S_\bullet = \mathbf{1}\mathbf{1}^\top - I - 2A_\bullet, \quad X_\bullet = I + \frac{1}{5}S_\bullet. \quad (2)$$

Exact finite certificate lemma. *The following identities hold exactly.*

1. Both graphs are $\text{SRG}(26, 10, 3, 4)$ and $S_c^2 = S_n^2 = 25I$.
2. Up to replacing z by $-z$, the set $Z_c = \{z \in \{\pm 1\}^{26} : S_c z = 5z\}$ has 130 elements and

$$5 \sum_{z \in Z_c} z z^\top = 130(5I + S_c). \quad (3)$$

3. The analogous set Z_n has 14 elements. The fourteen vectors

$$F(z) = (1, (z_i z_j)_{0 \leq i < j < 26}) \in \mathbb{R}^{326}$$

are linearly independent. The unique coefficients satisfying

$$(1, ((S_n)_{ij}/5)_{i < j}) = \sum_{z \in Z_n} w_z F(z) \quad (4)$$

are, in a deterministic lexicographic ordering,

$$-\frac{3}{10}, \underbrace{\frac{1}{10}, \dots, \frac{1}{10}}_{13 \text{ times}}. \quad (5)$$

Verification of the lemma. The accompanying script `code/verify_counterexample.py` decodes the two embedded graph6 strings and checks the degree and common-neighbor conditions, as well as $S^2 = 25I$, with integer arithmetic.

For each S , it computes a rational basis of $\ker(S - 5I)$ and selects thirteen pivot coordinates on which restriction is invertible. Every sign vector in the eigenspace is therefore uniquely determined by one of the 2^{13} sign assignments on those coordinates. A common-denominator integer reconstruction exhausts these assignments, proving the counts 130 and 14. It verifies (3) entry by entry. In the second case it computes the exact rational rank of the 14×326 feature matrix, solves a nonsingular 14×14 subsystem, obtains (5), and verifies all 326 equations in (4). The deterministic output is saved as `code/exact_certificate.json`. $\square$

Proof of the theorem. Because $S_\bullet^2 = 25I$ and $\text{tr}(S_\bullet) = 0$, the eigenvalues of each $S_\bullet$ are $+5$ and -5 , each with multiplicity 13. Thus

$$P_{\bullet,+} = \frac{1}{2} \left(I + \frac{1}{5} S_\bullet \right)$$

is the rank-13 orthogonal projector onto the $+5$ eigenspace and $X_\bullet = 2P_{\bullet,+}$. In particular, $X_\bullet$ is positive semidefinite, has diagonal one, has all off-diagonal entries of modulus $1/5$, and has nonzero eigenvalue $2 = 26/13$. It is therefore the Gram matrix of a real ETF of 26 vectors in $\mathbb{R}^{13}$.

Dividing (3) by 650 gives

$$X_c = \frac{1}{130} \sum_{z \in Z_c} z z^\top,$$

so $X_c \in \mathcal{C}^{26} = \mathcal{E}_{26}^{26}$. Its largest hierarchy degree is 26.

Suppose instead that $X_n \in \mathcal{C}^{26}$. Then

$$X_n = \sum_{z \in \{\pm 1\}^{26}} \lambda_z z z^\top, \quad \lambda_z \geq 0, \quad \sum_z \lambda_z = 1. \quad (6)$$

If $u \in \ker X_n$ then

$$0 = u^\top X_n u = \sum_z \lambda_z (u^\top z)^2.$$

Hence every z with $\lambda_z > 0$ is orthogonal to $\ker X_n$, so it belongs to $\text{ran } X_n = \ker(S_n - 5I)$. Up to sign, (6) can therefore use only the fourteen lines Z_n in the lemma. Taking diagonal and off-diagonal entries in (6) produces exactly (4), so linear independence forces the unique weights (5). One is negative, contradicting (6). Thus $X_n \notin \mathcal{C}^{26} = \mathcal{E}_{26}^{26}$.

The hierarchy levels are even, so the largest degree for X_n is at most 24, whereas that for X_c is 26. Both frames nevertheless have the same $(N, r) = (26, 13)$, proving the claim. $\square$

Scope and limitations. This completely answers the yes/no parameter-dependence sentence of Question 4.1 in the negative. It does not determine the exact largest degree for X_n ; the source’s degree-4 criterion gives $X_n \in \mathcal{E}_4^{26}$ because $26 < 13 \cdot 14/2$, so its depth lies between 4 and 24. Nor does the result classify all real ETFs or all regular two-graph switching classes.

Novelty search. The run indexes had no exact or core-question match. Bounded searches through 2026-08-13 used the exact Question 4.1 wording and the terms “ETF generalized elliptope”, “regular two-graph cut polytope”, “SRG(26,10,3,4) cut polytope”, and “(26,13) equiangular tight frame cut polytope”. They found the cited ETF/SOS papers, Laurent’s hierarchy lower bound, the graph catalogues, and general (26, 13) ETF constructions, but no same-parameter counterexample or the exact affine certificate (5).

Human-review recommendation. Check first the kernel-support implication following (6), then independently rerun the exact verifier and inspect its pivot-coordinate exhaustiveness and the 326-equation affine certificate. Subject to those checks, the counterexample is ready for expert review.

References.

- A. S. Bandeira and D. Kunisky, *Sum-of-Squares Optimization and the Sparsity Structure of Equiangular Tight Frames*, arXiv:1901.10697 (2019).
- A. S. Bandeira and D. Kunisky, *A Gramian Description of the Degree 4 Generalized Elliptope*, arXiv:1812.11583 (2018).
- M. Laurent, *Lower Bound for the Number of Iterations in Semidefinite Hierarchies for the Cut Polytope*, Math. Oper. Res. 28 (2003), 871–883.
- B. D. McKay, *Combinatorial Data: Strongly Regular Graphs*, <https://users.cecs.anu.edu.au/~bdkm/data/graphs.html>, catalogue file `sr261034.g6`, accessed 2026-08-13.